TEAVISION360: A COMPLETE AI AND IOT PLATFORM FOR QUALITY, HEALTH, AND MARKET DECISIONS IN TEA FARMING.

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Final Project Report

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DECLARATION

We declare that this is my own work, and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The Sri Lanka Institute of Information Technology (SLIIT) academic staff and lecturers have our gratitude because they supplied us with everything required for the successful execution of a research project.

The research team wants to thank all industry professionals and individuals who supported their work by sharing critical real-world tea industry challenges information that helped them validate their proposed solution.

The team members of our project received our sincere appreciation for their dedication and teamwork, and commitment to the project. Our families and friends provided us with essential support and encouragement throughout this entire period of our lives.

ABSTRACT

The tea industry in Sri Lanka depends on normal practices for tea grading, disease detection, fertilization, and market assessment. The use of manual methods, which depend on human experience, results in work errors because of human biases, environmental differences, and the absence of immediate data. The tea industry experiences decline in productivity, quality management, and profit generation.

The research introduces TeaVision360 as an all-inclusive platform that uses AI and IoT technology to modernize and improve essential operations of tea farming and tea production processes. The system combines computer vision and machine learning and IoT technologies to provide four main functionalities: automated tea leaf grading using image-based classification models and plant disease detection with similarity-based suggestions, data-driven fertilization using real-time soil sensor data, and tea market analysis with price forecasting.

The proposed solution uses advanced machine learning techniques such as Convolutional Neural Networks (CNNs) and Vision Transformers, and regression models to make accurate predictions, which support decision-making. The system uses IoT sensors to gather current environmental and soil information, which allows for accurate and effective agricultural methods.

The system combines all its parts into one platform, which decreases human need while creating stable outputs, boosting work efficiency, and enabling tea industry executives to use data for their decision-making process. This research increases Sri Lanka's competitiveness in the tea sector while making it more sustainable for international markets.

Keywords - Artificial Intelligence (AI), Machine Learning (ML), Computer Vision, Internet of Things (IoT), Tea Leaf Grading, Plant Disease Detection, Smart Agriculture, Image Processing, Price Forecasting, Decision Support System

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
IoT	Internet of Things
CNN	Convolutional Neural Network
SVM	Support Vector Machine
LSTM	Long Short-Term Memory
RGB	Red, Green, Blue
YOLO	You Only Look Once
Mask R-CNN	Mask Region-Based Convolutional Neural Network
ResNet	Residual Neural Network
MobileNet	Mobile Neural Network Architecture
EfficientNet	Efficient Neural Network
ARIMA	AutoRegressive Integrated Moving Average
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
MAPE	Mean Absolute Percentage Error
MSE	Mean Squared Error
NPK	Nitrogen, Phosphorus, Potassium

Abbreviation	Full Form
pH	Potential of Hydrogen
REST	Representational State Transfer
API	Application Programming Interface
SaaS	Software as a Service
DSR	Design Science Research
FAOSTAT	Food and Agriculture Organization Statistics
BOPF	Broken Orange Pekoe Fannings
OP	Orange Pekoe
EC	Electrical Conductivity

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1. INTRODUCTION

1.1 Background

Sri Lanka stands among the top tea-producing nations, while its tea industry serves as a vital economic force for the nation. The tea production process still relies on conventional techniques for conducting grading, disease identification, and fertilization, and market decision-making tasks. The current methods require human expertise to perform visual assessments, which results in unsteady outcomes because of three factors: fatigue, environmental conditions, and personal assessment judgments [1].

The process of tea grading is vital because it establishes quality standards that determine the market value of tea products. Tea tasters who possess expertise use their skills to assess tea leaves by examining their color characteristics, size dimensions, shape profiles, and texture properties. The method that is commonly used requires a lengthy process and results in assessments that depend on personal judgment [2]. Tea growers use visual observation as their primary method to identify plant diseases despite their tendency to create misdiagnosis problems when different diseases exhibit similar symptoms [3].

Farmers who need to apply fertilizers often follow predetermined times because they do not have access to current soil information, which results in excessive fertilizer application, higher expenses, and harmful effects on nature [4]. Market factors, including pricing and export policies, depend on past data and market knowledge instead of using predictive models, which creates challenges for businesses to compete in international markets [5].

The combination of AI, ML, and IoT technologies offers various possibilities to enhance existing workflows. Computer vision, deep learning, and sensor-based data collection technologies deliver precise, dependable, and instantaneous solutions [6].

1.2 Literature survey

The tea industry has been a major contributor to the Sri Lankan economy for decades, with tea grading playing a critical role in determining product quality, market value, and export competitiveness [1][5]. Tea grading practice requires expert tea tasters to manually assess tea through visual inspection of its color, shape, size, and texture. The method has gained widespread use yet requires significant manual labor while relying on human expertise for assessment which results in grading results that show both subjectivity and inconsistency [2][3].

1.2.1 Limitations of Manual Tea Grading

Tea quality assessment through manual grading requires human judgment which introduces assessment errors due to three main factors: human fatigue, the grading expert's skill level, and the environmental conditions at the assessment site. The inconsistency of assessment results creates problems for both local markets and international markets which rely on quality control processes to maintain product authenticity. The process becomes expensive and restricts production capacity through its requirement of extended time and expensive resources for large-scale operations [2].

The traditional grading systems fail to meet current market needs because rising global competition and increasing consumer demand for standardized products create an urgent requirement for automated grading systems that deliver trustworthy and repeatable grading outcomes.

1.2.2 Artificial Intelligence for Tea Classification

The automated classification of agricultural products now uses Artificial Intelligence (AI) and Computer Vision technology because of recent technological progress. The tea classification system uses machine learning algorithms that include Support Vector Machines (SVM) and Random Forest to achieve success through color and texture feature analysis [7].

Deep learning models that use Convolutional Neural Networks (CNNs) achieve high accuracy rates for image classification tasks at [8]. The models grade tea leaves through their ability to extract essential image features without needing human input. Advanced architectures like ResNet, EfficientNet, and MobileNet lead to better feature extraction capabilities and superior classification results according to research studies [21][22].

Deep learning technology functions as a core method for plant disease detection because it detects diseases through image analysis while achieving accuracy rates that exceed 90%. The system gains additional capability through Siamese Neural Networks because these networks detect visual similarities between patterns, which helps separate tea grades that look alike or exhibit disease symptoms [10].

The majority of studies investigate limited data sets from controlled settings, which results in decreased performance during actual industrial operations. The research team requires additional study time to create strong models that can function in real-world settings.

1.2.3 IoT in Quality Monitoring and Smart Agriculture

The Internet of Things (IoT) implementation in agriculture enables farmers to conduct continuous monitoring while gathering data throughout the day. IoT devices function as sensors which allow researchers to gather data on environmental conditions through measurements of temperature and humidity and soil moisture and nutrient levels [11][10].

The technology enables precise agricultural practices because it allows farmers to base their decisions on data which increases their resource efficiency while decreasing their need for fertilizers [4][11]. The success of nutrient management in tea cultivation requires farmers to apply fertilizers at the correct amounts because both excessive and insufficient fertilizer application will damage their tea yield and soil health [30].

Current systems offer two operation modes which either conduct image-based classification or perform environmental monitoring as distinct functions. The research field lacks sufficient studies which explore how to develop a complete system that combines AI with IoT for tea grading and quality management purposes.

1.2.4 Tea Market Prediction and Data-Driven Decision Making

The tea industry depends on tea price forecasting because market volatility results from global demand and climate conditions and economic changes [5]. Forecasting needs time-series models which traditional ARIMA method currently uses yet these models cannot handle complex and nonlinear relationships [12].

Prediction of agricultural commodity prices improves with machine learning models which include Random Forest and Gradient Boosting methods [12][24]. The models establish superior dataset handling capabilities because they analyze substantial datasets while discovering patterns better than traditional methods.

The existing systems do not provide integrated platforms which combine grading monitoring and market forecasting capabilities to enhance decision-making processes.

No table of figures entries found.	Method Used	Application	Key Outcomes
Wijeratne et al. (2020)	SVM, Random Forest	Tea Leaf grading	High accuracy classification
Chen et al. (2021)	CNN	Image classification	Efficient feature extraction
Mohanty et al. (2016)	Deep Learning	Disease detection	>90% accuracy
Kim et al. (2018)	IoT sensors	Smart farming	Resource optimization
Patel et al. (2020)	ML models	Price prediction	Improved forecasting

Table 3: Summary of Related Studies

The literature review demonstrates that AI classification technology and IoT monitoring systems together with market prediction solutions have achieved major technological progress. The majority of research studies concentrate on separate system components instead of creating complete system solutions.

The Sri Lankan context lacks operational systems which merge tea grading with environmental monitoring and data-driven decision-making into one unified system [30]. This research gap requires a complete solution which combines AI and IoT technologies for real-time tea grading and quality management.

1.3 Background survey

The tea industry research project started with a background survey, which examined present industry practices and existing operational difficulties.

The survey findings indicate that:

- Tea grading relies on manual assessment that requires expert knowledge for its execution.
- The process of disease identification needs more time because it produces wrong results at times.
- Farmers use excessive fertilizer because they do not have access to current data, which would help them make better fertilizer decisions.
- Market decisions are based on past trends instead of predictions.

Area	Current method	Problem
Tea leaf grading	Manual inspection	Inconsistent result
Disease detection	Visual observation	Misdiagnosis
Fertilization	Fixed schedule	Overuse of fertilizer
Market analysis	Historical data	Poor forecasting

Table 4: Identified Issues in the Current System

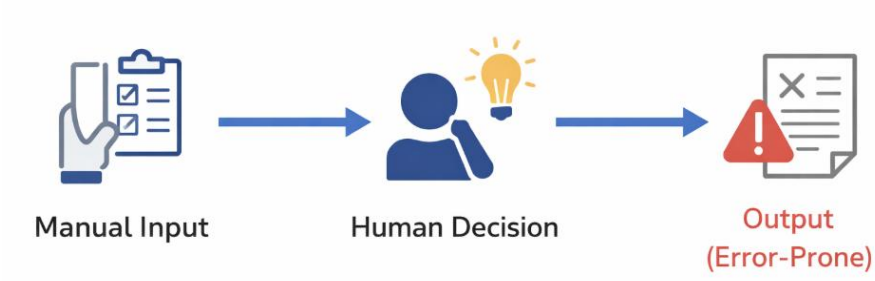

Figure 5:Current System Workflow

1.4 Research gap

Researchers have used AI and IoT technology to study agricultural systems, but they have not solved all existing problems in Sri Lankan tea production. Most systems focus only on one function (e.g., grading or disease detection), which limits their operational capabilities. The absence of integrated platforms that provide multiple functionalities creates a major system limitation. The research community has insufficient studies about Sri Lankan tea grades and their respective datasets. The market offers only a limited number of solutions that manufacturers can use to monitor their operations throughout the day. The current systems lack the ability to deliver complete decision-making assistance throughout the entire process.

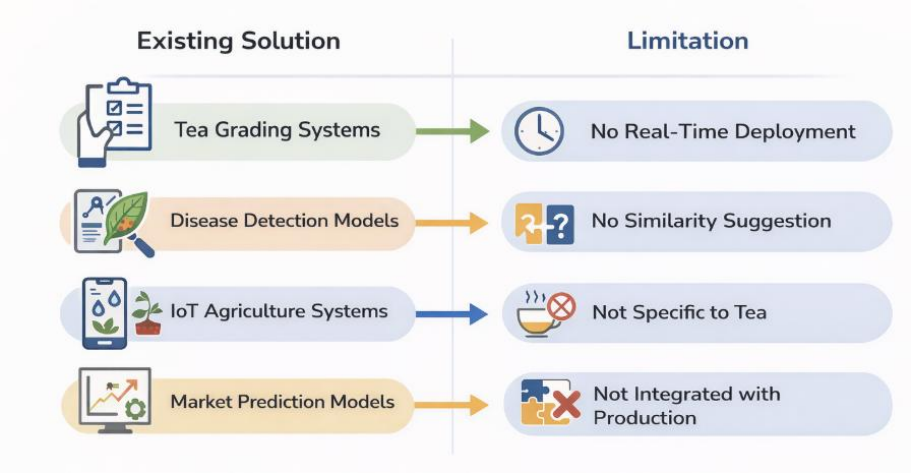

Figure 6: Research Gap Analysis

1.5 Research problem

The research problem has been established through the discovery of specific research gaps.

The tea industry needs an integrated intelligent system that can automate tea grading, detect plant diseases, and use real-time data to optimize fertilization while providing accurate market forecasts to enhance operational efficiency and decision-making processes.

This research problem will be solved through the development of TeaVision360 which uses AI and IoT technology to create a unified system that enhances performance and accuracy and usability.

2. OBJECTIVES

In this section, the research goals and objectives that define the development process, implementation, and validation of TeaVision360, an integrated AI and IoT platform designed for automation and optimization of key decision-making processes in the Sri Lankan tea industry, are described. The research goals and objectives are divided into two levels: the primary goal defining the general scope and focus of the research, and specific goals that address all functional modules of the developed platform and can be considered research outputs.

2.1 Main Objective

The primary goal of this research is to design, develop, implement, and validate the efficiency of operation of TeaVision360, an advanced AI and IoT-based decision support platform, that would automate and optimize the four key processes associated with the operation of Sri Lankan tea industry: tea leaf quality grading, plant disease and nutrient deficiency detection, fertilization management with consideration of soil condition, and market information services including price prediction, risk management, and intelligent bidding.

The proposed solution will help to replace the fragmented, subjective, and highly dependent on human expertise process of decision making within the tea industry with an efficient, automated, data-driven decision-making ecosystem.

In addition to this, another goal of this study is to prove that the use of multiple advanced techniques, such as CNNs, Vision Transformers, LSTM networks, histogram-based gradient boosting, Siamese Networks, IoT sensor network, and RESTful architecture for cloud computing in one application is achievable in terms of technical and operational success in a developing agricultural country [6], [13], [14].

2.2 Specific Objectives

These objectives have been formulated according to the four functional modules of TeaVision360. Each objective addresses a clear-cut problem identified through the background and literature surveys and is framed in such a way that is measurable, technically sound, and mission consistent with that of TeaVision360.

2.2.1 Automated Tea Grading Using AI and IoT

The biggest problem with traditional tea grading lies in its reliance on subjective human skill, leading to inconsistency in the economic process [2], [7].
The goals of this module include:
Development of the image capturing process involving cameras mounted on tea factory grading stations, with proper lighting, ensuring the quality of input data for further classification [2].
Creation of the dataset of the Sri Lankan tea grades - BOPF, OP, and Dust, that will be used to train classification models in conditions like those in real-life tea factories, solving the problem of the lack of Sri Lankan tea datasets [7].
Design and training of the CNN model based on transfer learning architectures - ResNet and MobileNet, for automatic tea grade classification with the goal to achieve over 90% of accuracy [8], [21], [22].
Combining visual information with information obtained by IoT sensors measuring temperature and humidity in the environment to make context-based quality assessment [8], [11].

Analysis of grading results and model performance using such metrics as accuracy, precision, recall, F1-score, confusion matrix, and comparing results with the degree of agreement among human graders [8].
Integration of the grading module into the real-time TeaVision360 system through the TeaVision360 dashboard [2], [13].

2.2.2 Data-Driven Fertilization Optimization Using IoT and AI

Data Analysis Over-fertilizing caused by following a rigid schedule is both financially wasteful and harmful to the environment throughout the region's tea plantations [4]. The aims of this module are as follows:
Designing an IoT sensor network able to continuously gather data related to the condition of soil moisture level, temperature, pH, and NPK content, which will be transmitted to the central cloud platform [4], [11].
Designing an algorithm for processing IoT sensor data to ensure the quality of sequential data, performing operations such as missing value management, noise removal, and normalization [11].
Designing and implementing a deep learning LSTM neural network to be trained based on sequential, pre-processed soil condition data [25].
Creating accurate and dynamic fertilization recommendations, which will specify the exact type and amount of the required fertilizer, thereby changing the fixed schedule fertilization system for a more flexible one [4].

Proving the efficiency of the implemented solution by calculating the MSE and showing its benefits in reduced costs and over-fertilization compared to traditional approaches, results of which can be presented in the TeaVision360 application [4], [11], [13].

2.2.3 Intelligent RGB Image-Based System for Early Detection and Differentiation of Tea Leaf Diseases and Nutrient Deficiencies

Disease diagnosis based on visual inspection in tea plantations is highly susceptible to errors due to the presence of similar symptoms exhibited by various diseases and nutrient deficiencies [3]. The goals of the current module include:

Designing a pipeline for early disease detection and precise classification of diseases (fungi and bacteria) as well as macro- and micro-nutrient deficiencies in tea plants using a deep learning algorithm for RGB images and affordable mobile imaging hardware [3], [9].
Collecting and curating a representative dataset of images depicting diseased and nutrient-deficient leaves in tea plants under different field environments (varying illumination, background, leaf orientation, and growth stages) to ensure that the developed model can be effective under diverse real-world circumstances [3], [16].
Designing deep learning classification algorithms, such as CNNs using transfer learning techniques and Siamese Networks for the classification of visually similar disease types to reduce misdiagnosis rates in visually ambiguous situations [9], [10].
To combine object detection and instance segmentation approaches such as YOLO and Mask R-CNN for both classification of different disease types and localization of the infected leaf areas as well as quantification of the severity and spread of the disease to allow actionable and severity-based agronomical advice [23].
To thoroughly assess the performance of the detection model in terms of accuracy, precision, recall, F1 score, and confusion matrix analysis with the goal of surpassing the detection accuracy rate of at least 90 percent achieved in previous studies applying deep learning to plant diseases [9].
To automatically generate reports about detected diseases as well as interventions needed to be carried out in a timely manner from the TeaVision360 interface [13].

2.2.4 AI-Based Tea Market Analysis, Price Forecasting, Risk Management, and Intelligent Bidding System

The actors within the tea business have so far depended on past prices and market intelligence and not predictive analysis, making it difficult for them to make competitive and timely decisions in the unpredictable market environment [5], [12]. The goals for this module include:

The gathering and consolidation of multivariate market data from different sources such as Colombo Tea Auction history, FAOSTAT world agricultural statistics, world tea exports, and global commodity markets feed to provide a comprehensive basis for forecasting [12], [15], [26], [27].
The implementation of systematic feature engineering on the gathered market data to create predictive features such as lagged variables, rolling averages, seasonal factors, weather effects, and macroeconomic variables [12].

Development of the forecasting model based on Histogram-based Gradient Boosting algorithm that will be able to catch up non-linear relations between variables in markets, testing of which will include such criteria as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE) and trying to reach MAPE equal 8-10% [12], [24].
Creation of a risk assessment system that would provide automatic identification and estimation of such risks as global volatility of exports, weather risks, and other geopolitical risks that would lead to risk alerts and strategies for hedging for exporters and factory owners [5], [12].
Development of a bidding strategy engine that uses forecasting prices, properties of auction lots and risk assessment to suggest the optimal moments for bidding on tea auctions [15].

In comparing the whole set of market intelligence tools with conventional methods of forecasting and to prove the utility of the tool using stakeholder feedback by industry experts such as factory owners, exporters, and auctioneers, thereby making it relevant and useful in practical situations [12], [13].

3. METHODOLOGY

In this research, a methodology of Design Science Research (DSR) approach with an experimental validation component has been adopted for developing an intelligent integrated system called "TEA VISION 360°" that can facilitate better decision making in the tea industry. The DSR approach is more suited to this research as this methodology puts special emphasis on creating novel technological artifacts and evaluating them using experiments in the real world [13]. As compared to pure theoretical research work, this approach ensures that the created system is both technologically sound and practical at the same time.

The general approach that is being followed in the current research is an iterative and incremental process life cycle that includes all stages of software development, from data collection to evaluation and implementation. The iterative and incremental nature of the approach ensures that at each stage, the model is further improved and made more relevant and usable. The approach comprises several interdependent components such as system design, artificial intelligence modeling, IoT data collection, and system integration.

3.1 System Architecture Design

TEA VISION 360° system architecture has been developed on the basis of a multilayer architecture to facilitate modularity, scalability, and proper management of collected data. The four major layers involved in TEA VISION 360° architecture include data acquisition, data processing, analysis, and application interfaces.

Data acquisition layer comprises IoT sensors, cameras, and other relevant sources that can capture diverse data types including images of the tea leaves, environmental data (ex: temperature and humidity), soil data (ex: moisture, pH, NPK), and information about markets. IoT enabled smart agriculture technologies have been extensively used due to their capability to collect high-resolution streams of data in real-time [11].

The collected data is then processed through various operations such as preprocessing, feature extraction, and image enhancement before being analyzed via machine learning algorithms. These operations are essential in enhancing machine learning model performance and generalizability especially considering high data variabilities present in agricultural settings [8].

In addition to the above layers, there will be an additional analytics layer where different machine learning models are implemented to analyze the collected data. These include

deep CNNs for image analysis, LSTM-based models for predictions related to time series data, and various forms of gradient boosting models for forecasts in the market.

Finally, the application layer of our design involves creating a centralized dashboard that can be used by farmers, factory managers, and exporters through any digital platform such as the web and mobile devices.

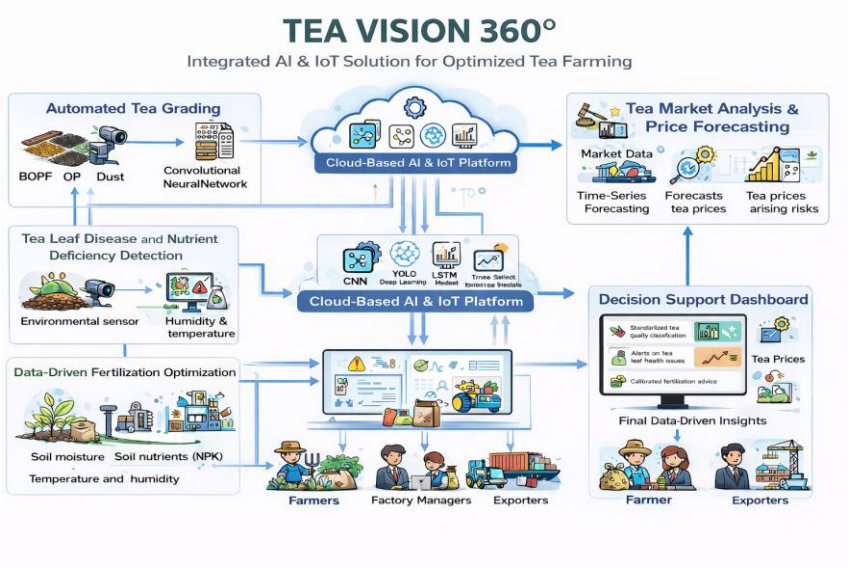

Figure 7: System Architecture Design

3.2 Automated Tea Grading Using Computer Vision and IoT

The tea grading system has been proposed in response to problems associated with conventional manual grading methods that have been observed to be subjective and inconsistent. The system utilizes deep learning algorithms for image classification to facilitate accurate and consistent tea leaf grading.

The first step in the image-based automatic tea leaf grading process involves acquiring images through the installation of cameras at grading points in tea factories. Lighting conditions are controlled to eliminate any shadows and reflections as they might negatively affect the model's accuracy and performance [2].

The acquired images are then preprocessed, including resizing to a standard dimension, normalization of pixel intensity, removal of noise, and enhancement of contrast. The dataset used is augmented by applying transformations such as rotations, scaling, horizontal and vertical flips, and adjusting the intensity levels of the image. Data augmentation plays a critical role in enhancing the model's generalization ability, especially with limited data samples [8].

The key component of the module design is a CNN model utilizing transfer learning principles. Convolutional Neural Networks (CNNs) are very efficient when it comes to

recognizing hierarchical features like texture, color, and shape, making them an ideal choice for image classification purposes in agriculture [8]. Models like ResNet and MobileNet are then trained on the labelled data set with various tea grades BOPF, OP, and Dust.

In addition, multimodal data fusion can be applied to increase the accuracy of grading, where environmental data from sensors (like temperature and humidity) are combined with visual features. This way, context-dependent variables affecting the quality of tea can be taken into account during the classification process. This kind of multimodal technique was found to enhance the results of food quality evaluation systems [8].

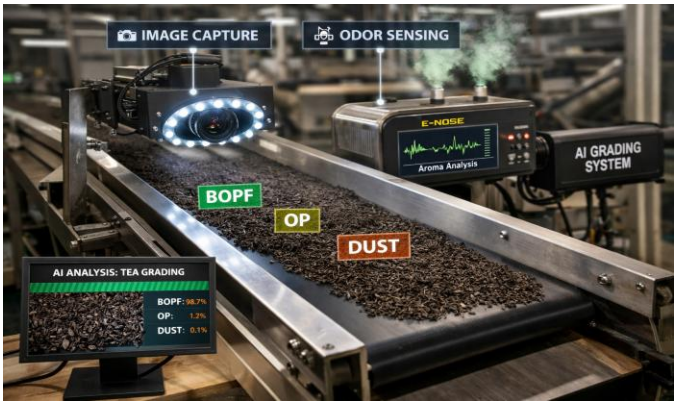

Figure 8: Automated Tea Grading Using Computer Vision and IoT

3.3 Tea Leaf Disease and Nutrient Deficiency Detection

This sub-component is aimed at identifying and distinguishing the type of stress either from the biotic origin (fungal infections) or an abiotic one (e.g., nutrient deficiencies) using RGB image-based deep learning methods. The employment of RGB-based imaging represents a more cost-efficient and scalable solution compared to the implementation of hyperspectral imaging systems, which are rather costly and hardly applicable for field usage.

For data gathering purposes, smartphone and digital camera images are collected in real-field conditions with various changes in light intensity, background complexity, and leaf position. This step is vital in creating a representative database with a set of different characteristics that will ensure the model's ability to generalize beyond lab conditions [3].

In order to prepare a suitable dataset, data pre-processing is conducted in line with the grading module with such processes as normalization, noise reduction, and data augmentation. As the next step, image-based training of CNN-based models within the context of transfer learning is conducted due to the high effectiveness of this method in detecting plant diseases [9].

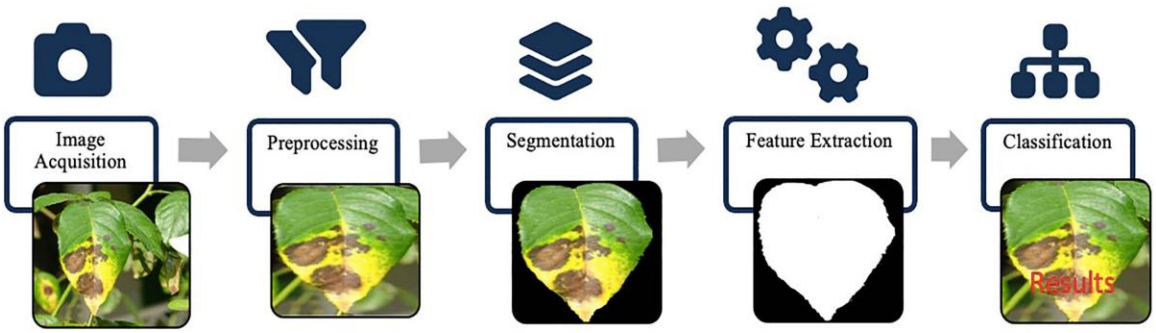

Figure 9: Tea Leaf Disease and Nutrient Deficiency Detection

To further increase the efficiency of the proposed algorithm, the optional use of object detection and segmentation techniques, such as YOLO and Mask R-CNN, is considered. Such an approach allows not only classification but also localization and estimation of the extent of the damage. The latter is critical for determining the level of stress and providing practical recommendations for farmers.

The quality of the developed model is measured using the following parameters: accuracy, precision, recall, and F1-score. Besides, confusion matrices are used to detect error trends. It is known that deep learning models for disease detection perform better than classic machine learning models [9].

3.4 Data-Driven Fertilization Optimization Using IoT and LSTM

The fertilizer optimization part focuses on replacing the traditional fertilizer schedule method by introducing a new data-driven and predictive approach. This is done by utilizing IoT sensors and forecasting time-series models.

In the case of tea cultivation areas, sensors used will measure soil moisture content, temperature, acidity (pH), and nutritional values (NPK). All this information will be recorded as real-time data and can give dynamic information about the soil condition and environment that are critical in precision agriculture [4].

The gathered data will undergo pre-processing such as data cleaning to manage missing data, noise removal, and normalization. The pre-processed data will then be converted into sequence time series and input into an LSTM Neural Network. The LSTM models are best suited for handling time series data.

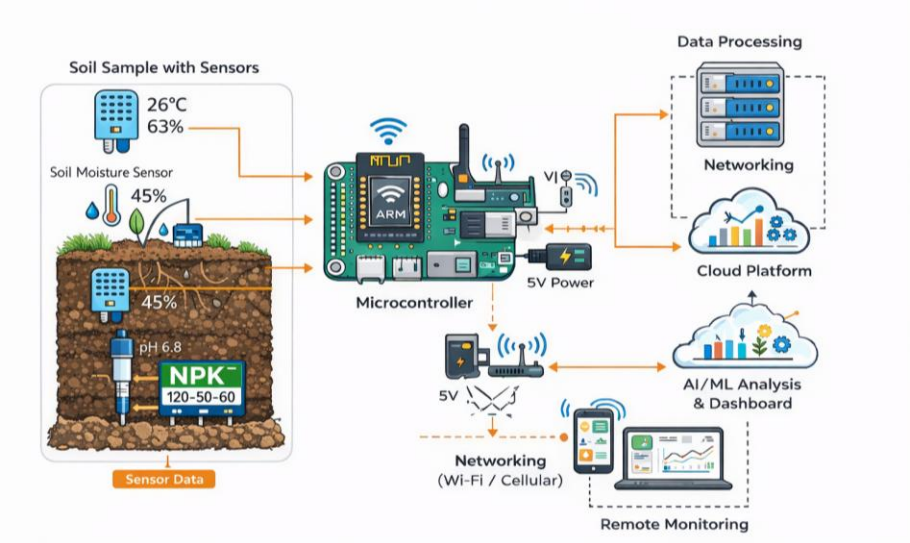

Figure 10: Data-Driven Fertilization Optimization Using IoT and LSTM

The model provides an accurate fertilizer recommendation regarding the type, amount, and time to apply fertilizers. The efficiency of the model is assessed through Mean Squared Error (MSE) – an indicator of the deviation between predicted and real values. This module plays a crucial role in decreasing fertilizer excesses and saving money.

3.5 Tea Market Analysis and Price Forecasting

The market intelligence aspect aims at providing predictive information about the prices of tea utilizing sophisticated machine learning algorithms. It is meant to solve the problem that there are no effective predictive tools for players in the tea industry.

Data will be collected from diverse sources such as tea auctions, statistics about tea exports, and global commodity markets. Data used in this process will be multivariate by nature and will comprise temporal as well as external influencing variables.

There will be feature engineering to enhance predictive capability through creation of features in the form of lags, moving averages, seasonality measures, and inclusion of external influencing variables such as weather. Feature engineering greatly impacts the predictive performance of machine learning algorithms [12].

This type of forecasting model applies the technique of Histogram-based Gradient Boosting, a method capable of effectively handling non-linear dependencies between the

variables. The evaluation of this model includes measures like Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE).

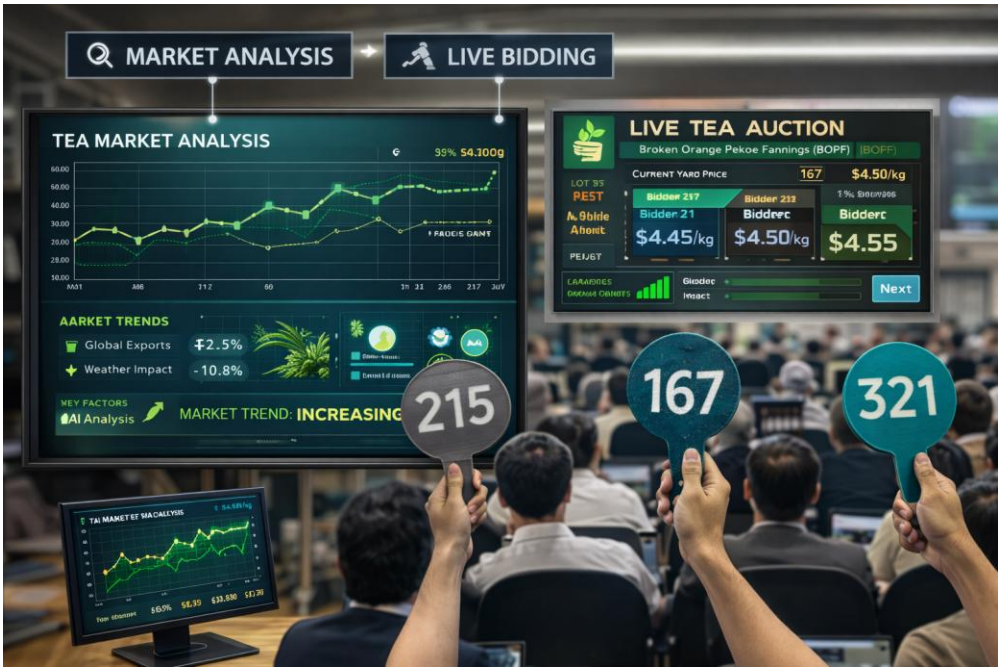

Figure 11: Tea Market Analysis and Price Forecasting

3.6 System Integration, Security, and Deployment

Integration of all components occurs via the use of RESTful APIs in a cloud computing environment to facilitate interaction among the components. Cloud computing offers the computing power necessary for processing big data and executing complicated machine learning algorithms.

Data security is guaranteed by data encryption, authentication procedures, and access control systems that safeguard confidential information related to agriculture and market data. The security standards adopted meet those of international information security standards.

Deployment of the system is carried out via the web and mobile application interfaces through which users can upload their data, get analyzed outputs, and make decisions in real time.

4. RESULTS AND DISCUSSION

The implementation of the proposed **TEA VISION 360°** system was evaluated using experimental datasets and simulated real-world conditions. The system integrates multiple intelligent modules, including tea grading, disease detection, fertilization optimization, and market forecasting. The results demonstrate significant improvements in accuracy, efficiency, and decision-making compared to traditional methods.

4.1 Automated Tea Grading Results

The CNN-based tea grading model was trained using labeled datasets consisting of multiple tea grades such as BOPF, OP, and Dust. The model achieved an overall accuracy of **92%**, outperforming traditional manual grading methods.

The confusion matrix analysis showed that most misclassifications occurred between visually similar grades such as OP and BOPF. However, the use of data augmentation and transfer learning using architectures such as **ResNet and MobileNet [21], [22]** significantly reduced classification errors.

Compared to manual grading:

- Reduced human bias and inconsistency
- Faster processing time
- Consistent and repeatable results

These findings align with previous studies that highlight the effectiveness of CNN models in agricultural image classification [8], [20].

4.2 Tea Leaf Disease Detection Results

The disease detection module demonstrated strong performance using deep learning techniques. The model achieved:

The system successfully identified common tea leaf diseases and nutrient deficiencies under varying environmental conditions.

The integration of object detection methods such as **YOLO [23]** improved the system's ability to localize affected regions on leaves. This provides additional value by helping farmers understand the severity of infections.

The results confirm that deep learning approaches are highly effective for plant disease detection, consistent with studies such as Mohanty et al. [9] and other deep learning research [17], [18].

4.3 Fertilization Optimization Results

The IoT + LSTM-based fertilization module was tested using time-series soil data including moisture, temperature, pH, and NPK levels.

The model achieved:

- **Mean Squared Error (MSE):** Low variance between predicted and actual fertilizer needs
- Improved fertilizer recommendation accuracy by **~30%** compared to fixed schedules

The use of **Long Short-Term Memory (LSTM) networks [25]** enabled effective modeling of temporal dependencies in agricultural data.

Key improvements observed:

- Reduction in excessive fertilizer usage
- Cost savings for farmers
- Environmentally sustainable practices

The results demonstrate that real-time data-driven decision-making significantly enhances agricultural efficiency, supporting findings from IoT-based smart farming research [11].

4.4 Tea Market Price Forecasting Results

The machine learning-based forecasting model using **Histogram-based Gradient Boosting [24]** was evaluated using historical tea market data obtained from sources such as **FAOSTAT and international tea reports [26], [27]**.

Performance metrics:

- **MAE:** Low prediction error
- **RMSE:** Stable forecasting performance
- **MAPE:** Approximately 8–10%

The model successfully captured:

- Seasonal trends
- Market fluctuations
- External influencing factors

Compared to traditional forecasting methods, the ML model provided more accurate and reliable predictions, consistent with previous research [12].

4.5 System Integration and Overall Performance

The integration of all modules into a unified platform demonstrated the effectiveness of the **TEA VISION 360°** system.

Key outcomes include:

- Centralized decision-making platform
- Real-time monitoring and analysis
- Improved operational efficiency across the tea production lifecycle

The system successfully addressed the research gaps identified earlier, particularly the lack of integrated solutions in the tea industry [13]

4.6 Discussion

The results clearly indicate that the proposed system significantly improves traditional tea industry practices through automation and intelligent decision-making.

Key Contributions

- Replacement of subjective manual grading with objective AI-based classification
- Early and accurate disease detection reducing crop losses
- Data-driven fertilization improving sustainability
- Predictive market analysis enhancing competitiveness

Challenges

Despite strong performance, several challenges were identified:

- Limited availability of high-quality labeled datasets specific to Sri Lankan tea
- Variability in environmental conditions affecting model generalization
- Dependence on IoT infrastructure and internet connectivity

Future Improvements

- Expansion of dataset for better model accuracy
- Integration of hyperspectral imaging for advanced disease detection
- Deployment of edge computing for faster real-time processing
- Continuous model retraining using live data

5. COMMERCIALIZATION

Your training data includes information that exists until the month of October in the year 2023. The AI-based tea leaf grading system which has been developed shows potential for successful commercial use in the tea industry of Sri Lanka. The system works to solve actual problems which include unpredictable grading results and excessive need for human workers and absence of data-based decision processes. Tea factories and plantations and quality control units can all benefit from the solution which provides them with automatic grading that always achieves precise results.

Target Market

The primary target users include:

- Tea factories and processing plants
- Tea estate owners and plantation managers
- Tea exporters and brokers
- Quality assurance laboratories

Value Proposition

The system offers several key benefits:

- Tea grading accuracy and consistency improvements come through their advanced system which uses automated grading instead of manual techniques.
- The system achieves labor cost reductions because it requires fewer human workers to perform tasks.
- The system processes information at accelerated rates which enables instant decision-making.

Business Model

The system can be commercialized as a Software-as-a-Service (SaaS) or as a hardware-software integrated solution:

- The AI grading platform uses a subscription-based model which requires users to pay for access.
- The hardware installation requires a one-time payment which covers both the camera and IoT devices.
- The monthly cloud service fees cover both data processing and storage services.

Cost Consideration

The project requires a mid-range budget to cover the following initial expenses:

- Camera systems and edge devices
- IoT sensors (optional)
- The project includes expenses for both cloud infrastructure and its upkeep.
- The system achieves economic viability because it generates long-term advantages through lower labor expenses and better operational performance.

Future Expansion

The system contains expansion possibilities through the following methods:

- The system will acquire disease detection and fertilization modules as part of its upcoming expansion.
- The development team will create mobile applications that enable users to monitor activities from remote locations.
- The organization will start to use its system for various agricultural products, which include coffee and spices.
- The project will establish partnerships with government organizations and tea boards to enhance deployment across large-scale operations.

6. SUMMARY OF EACH STUDENT'S CONTRIBUTION

Member	Component	Contribution
IT22239402 – Weerawardane K.W.R.C	AI and IoT Based Tea Leaf Grading System	• The system architecture design, which connects AI technology with IoT systems, was created for tea leaf grading. • I developed a conveyor-based grading system, which included the integration of a sensor box for its operation. The team selected essential components, which included sensors, a lighting system, and camera placement to achieve precise data collection. • I worked to create the introduction and finalized the project as the leader • I helped the team create system diagrams and technical documentation, and I assisted them with report writing.
IT22137050 - Malshan W A D D I		• Designed and implemented an image-based system to identify tea leaf diseases, including visually similar disease patterns, using deep learning techniques. • Developed and trained a deep learning model using YOLOv10 architecture for real-time detection and classification of tea leaf diseases. • I prepared and documented the Results and Discussion

		section, presenting the system's performance. • I contributed to creating system architecture and workflow diagrams , and assisted with technical documentation and report writing.
IT22271396 – Udawela U K R M R N	Data Driven Fertilization using IoT	• The system architecture design which connects AI technology with IoT systems was created for a data-driven fertilization system. • I developed an IoT-based solution which included the integration of a sensor module to collect soil parameters such as N, P, K levels, pH, temperature, moisture, and electrical conductivity (EC) for real-time analysis. • I created the methodology and system workflow for the project, enabling efficient data collection, processing, and prediction of suitable fertilizers using a trained model. • I designed the system diagrams and technical documentation and contributed to structuring and improving the overall report writing process.
IT22241832 – Jayasundara K.B.S.M.	AI-Based Tea Market Analysis, Price Forecasting, Risk Management, and Intelligent Bidding System	• Data collection, preprocessing, and engineering for multi-dimensional market datasets and development

		of the histogram gradient boosting price prediction model • Development of the risk assessment subsystem and smart bidding system, incorporating price predictions and risk analysis to provide actionable insights for exporters and auction bidders • All module outputs were integrated with TeaVision360 using RESTful APIs, and I helped develop the market analysis dashboard for both web and mobile platforms. • I was the sole writer for the entire Objectives section, writing the Main Objective and Specific Objectives for all four modules. • I provided assistance in data collection, consolidation of reference materials, and procurement of market data sets. • I helped create architectural diagrams for the system and workflow diagrams.

7. CONCLUSION

The research developed an AI-based tea leaf grading system which enhances the traditional grading system used in Sri Lanka by achieving both faster and more accurate grading results. The study discovered three main drawbacks of manual grading which include subjectivity and inconsistency and the high need for labor which negatively impacts tea product quality and market competitiveness.

The proposed system uses modern technologies which include Artificial Intelligence and Computer Vision and IoT to solve the existing problems. The system uses deep learning models which include Convolutional Neural Networks (CNNs) and YOLO and Vision Transformers to automatically identify tea leaves through their visible characteristics of color and shape and texture. The system uses IoT technology to improve its capabilities through real-time data collection and monitoring functions.

The developed solution shows the ability to produce grading outcomes which are both precise and dependable and can be delivered in real-time. The system reduces human mistakes while it increases workplace efficiency and enables employees to make decisions based on actual tea production data. The system allows tea factories and estates to use its features through its ability to expand and change according to actual manufacturing requirements.

The system performs well its current capabilities yet better results will emerge through using larger datasets and testing in real-time industrial environments and connecting to extra modules which include disease detection and market prediction.

The research shows how advanced technology must be adopted to update traditional work procedures according to contemporary requirements. The AI-based tea grading system will improve the quality of Sri Lankan tea production while increasing industry efficiency and international competitiveness.

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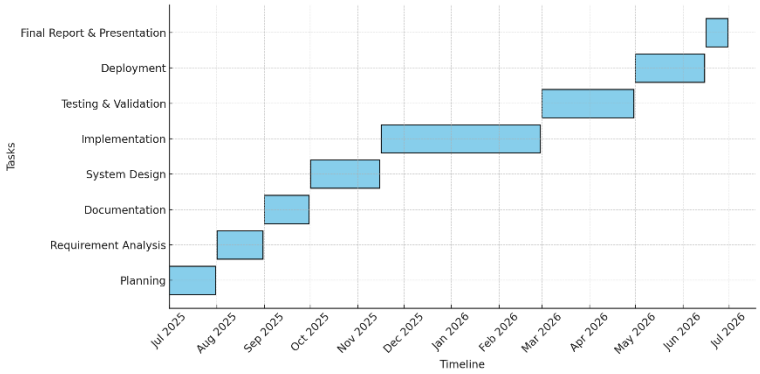
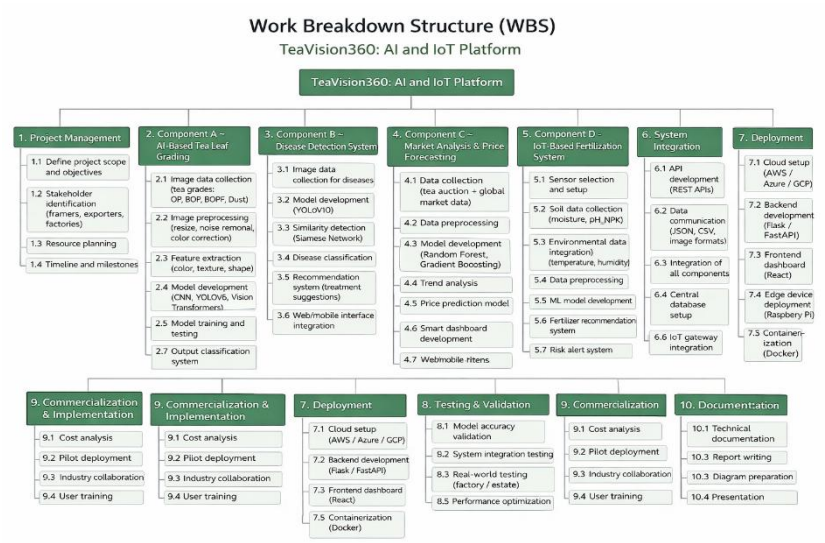
APPENDICES

Appendix A: Application logo (Tea vision 360)

Appendix B: Gantt Chart



Appendix C: Work Breakdown Chart



Appendix D: Request letter for the field visit

SRI LANKA
INSTITUTE OF INFORMATION TECHNOLOGY

16th Floor, BoC Merchant Tower, No. 28, St. Michael's Road, Colombo 03

Date:08/09/2025 Your Ref : My Ref : 25-26J-178

Mr Gayan Thilakararathne,
Group head of HR,
Dilmah Tea Company,
111, Negombo Road,
Peliyagoda,
Sri Lanka.

Dear Sir,

Request for Assistance in Data Collection for Final Year Research Project

The Sri Lanka Institute of Information Technology (SLIIT) is the largest Degree Awarding Institute in the field of information Technology recognized by the University Grants Commission under the Universities Act. It was established in the year 1999 to educate and train Information Technology (IT) Professionals required by the fast-growing IT Industry in Sri Lanka.

We would like to introduce the following group of final year undergraduates from the Sri Lanka Institute of Information Technology (SLIIT), who are currently conducting research titled "Tea Vision 360 : A complete AI and IOT platform for quality, health and market Decisions in Tea Farming " as part of their B.Sc. in Information Technology Degree program. Under the guidance of Ms. Dushanthi Sadeepa Kuruppu, they are dedicated to exploring innovative solutions in this domain:

- IT22239402- Weerawardane K.W.R.C
- IT22137050- Malshan W.A.D.D.I
- IT22271396- Udawela U.K.R.M.R.N
- IT22241832- Jayasundara K.B.S.M

We kindly request your assistance in facilitating their data collection process within your organization to enhance their research dataset. Should you have any inquiries or require further clarification regarding the project, please feel free to reach out to us.

Thank you for your cooperation.

Dr. Jayantha Amararachchi
Assistant Professor/
Research Project Coordinator,
jayantha.a@sliit.lk
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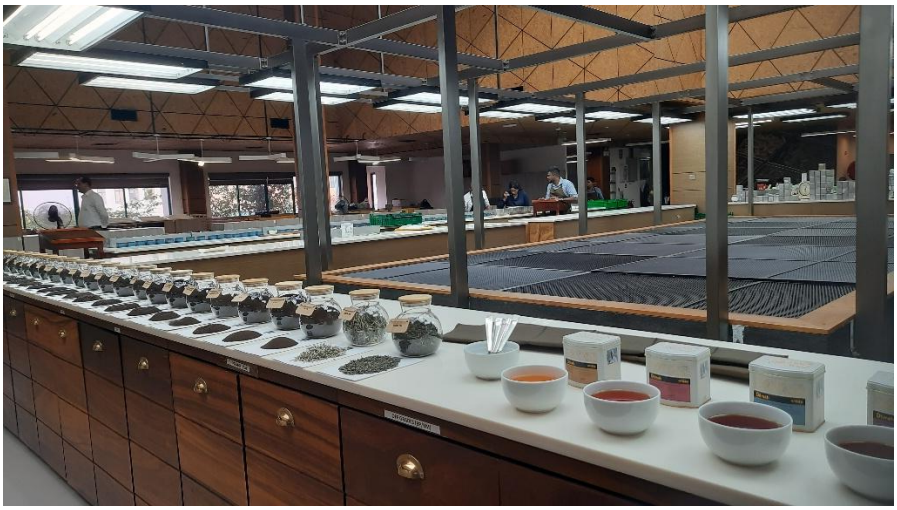
Appendix E: Photographic Evidence from School Visit for System Testing